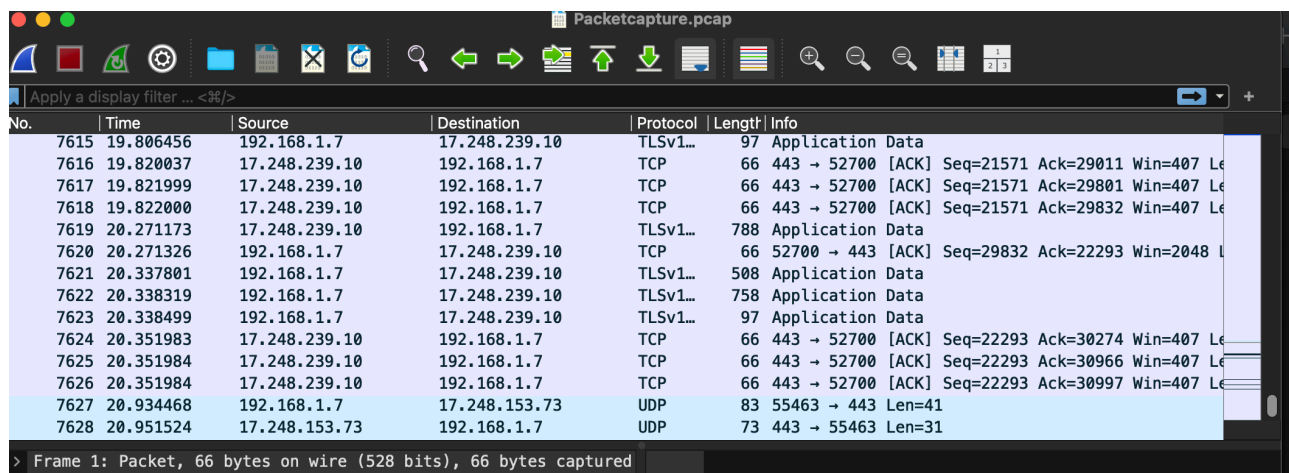


Module 2: Network + Network Security , Maintenance and Troubleshooting

Software and Hardware Tools

Task 1:



No.	Time	Source	Destination	Protocol	Length	Info
7615	19.806456	192.168.1.7	17.248.239.10	TLSv1...	97	Application Data
7616	19.820037	17.248.239.10	192.168.1.7	TCP	66	443 → 52700 [ACK] Seq=21571 Ack=29011 Win=407 Le
7617	19.821999	17.248.239.10	192.168.1.7	TCP	66	443 → 52700 [ACK] Seq=21571 Ack=29801 Win=407 Le
7618	19.822000	17.248.239.10	192.168.1.7	TCP	66	443 → 52700 [ACK] Seq=21571 Ack=29832 Win=407 Le
7619	20.271173	17.248.239.10	192.168.1.7	TLSv1...	788	Application Data
7620	20.271326	192.168.1.7	17.248.239.10	TCP	66	52700 → 443 [ACK] Seq=29832 Ack=22293 Win=2048 L
7621	20.337801	192.168.1.7	17.248.239.10	TLSv1...	508	Application Data
7622	20.338319	192.168.1.7	17.248.239.10	TLSv1...	758	Application Data
7623	20.338499	192.168.1.7	17.248.239.10	TLSv1...	97	Application Data
7624	20.351983	17.248.239.10	192.168.1.7	TCP	66	443 → 52700 [ACK] Seq=22293 Ack=30274 Win=407 Le
7625	20.351984	17.248.239.10	192.168.1.7	TCP	66	443 → 52700 [ACK] Seq=22293 Ack=30966 Win=407 Le
7626	20.351984	17.248.239.10	192.168.1.7	TCP	66	443 → 52700 [ACK] Seq=22293 Ack=30997 Win=407 Le
7627	20.934468	192.168.1.7	17.248.153.73	UDP	83	55463 → 443 Len=41
7628	20.951524	17.248.153.73	192.168.1.7	UDP	73	443 → 55463 Len=31

Top 3 protocols - UPD, TCP, ICMP

Task 2:

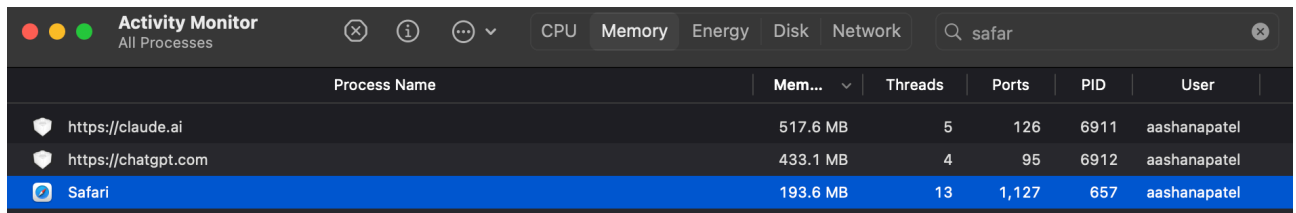
```
Last login: Mon Sep 21 19:23:17 on console
aashanapatel@192 ~ % ping www.spotify.com -c 4
PING atc.spotify.map.fastly.net (151.101.107.42): 56 data bytes
64 bytes from 151.101.107.42: icmp_seq=0 ttl=60 time=21.481 ms
64 bytes from 151.101.107.42: icmp_seq=1 ttl=60 time=22.825 ms
64 bytes from 151.101.107.42: icmp_seq=2 ttl=60 time=20.379 ms
64 bytes from 151.101.107.42: icmp_seq=3 ttl=60 time=27.403 ms

--- atc.spotify.map.fastly.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 20.379/23.022/27.403/2.674 ms
aashanapatel@192 ~ % ping www.zomato.com -c 4
PING e70929.dsca.akamaiedge.net (49.44.140.145): 56 data bytes
64 bytes from 49.44.140.145: icmp_seq=0 ttl=56 time=17.451 ms
64 bytes from 49.44.140.145: icmp_seq=1 ttl=56 time=22.245 ms
64 bytes from 49.44.140.145: icmp_seq=2 ttl=56 time=6.677 ms
64 bytes from 49.44.140.145: icmp_seq=3 ttl=56 time=6.427 ms

--- e70929.dsca.akamaiedge.net ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.427/13.200/22.245/6.861 ms
aashanapatel@192 ~ %
```

Zomato's response time is faster than Swiggy.

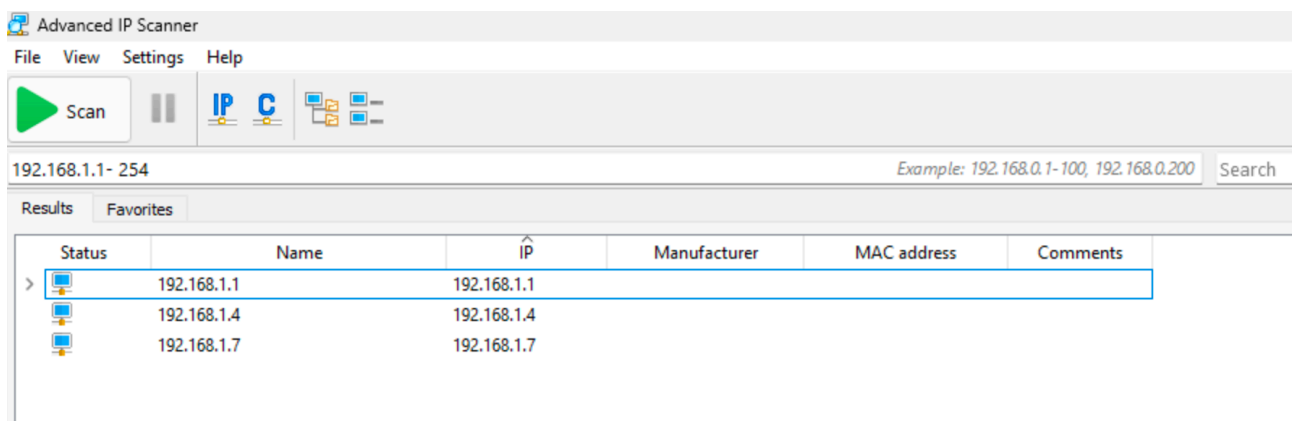
Task 3:



The screenshot shows the macOS Activity Monitor window with the 'Memory' tab selected. The search bar contains 'safari'. The table below lists the processes found.

Process Name	Mem...	Threads	Ports	PID	User
https://claude.ai	517.6 MB	5	126	6911	aashanapatel
https://chatgpt.com	433.1 MB	4	95	6912	aashanapatel
Safari	193.6 MB	13	1,127	657	aashanapatel

Task 4:



The screenshot shows the Advanced IP Scanner application. The scan range is set to 192.168.1.1-254. The results table shows three active IP addresses.

Status	Name	IP	Manufacturer	MAC address	Comments
Active		192.168.1.1			
Active		192.168.1.4			
Active		192.168.1.7			

Task 5:

Cable tester — use when you suspect a physical Ethernet/LAN cable is faulty (no connection, intermittent drops) and want to verify each wire pin is properly connected.

Crimping tool — use when making or repairing your own Ethernet cables, to attach RJ45 connectors to cable ends.

Multimeter — use to check electrical continuity or voltage on network hardware (e.g. verifying a PoE injector or router power supply is delivering correct voltage).

